Skewness_Kurtosis

kurtosis	-1.2	-0.60751	0.450765	0.052143	-1.08858	-0.470723	18.544273
skew	0.0	-0.132649	0.163639	0.244917	0.282308	0.313576	3.569747

Skewness and Kurtosis Analysis

Skewness

Skewness is used to measure the symmetry of the data distribution.
If the skewness value is close to 0, the data is considered balanced and normally distributed.

In this placement dataset, most skewness values are near 0 such as:
-0.13, 0.16, 0.24, 0.28, and 0.31.

This indicates that:

- The dataset is fairly balanced.
- There is no major left or right skewness.
- The data distribution is close to normal distribution.
- The dataset is suitable for analysis and model building.

Kurtosis

Kurtosis is used to identify the presence of extreme values or outliers in the dataset.

In this dataset:

- Most kurtosis values lie between -1 and 1, which indicates a normal distribution without major outliers.
- However, one kurtosis value is 18.54, which is very high.

Skewness_Kurtosis

This indicates that:

- One feature contains extreme outliers.
- Some values are unusually higher or lower compared to the remaining data.
- That feature may require preprocessing or outlier treatment before training the model.

Difference Between Skewness and Kurtosis

Skewness	Kurtosis
Measures symmetry of data	Measures of the presence of outliers
Indicates left or right distribution	Indicates extreme peak or tail values
Value near 0 means balanced data	High value means more outliers

Conclusion

The placement dataset is mostly well-balanced because the skewness values are close to zero. Most kurtosis values are also normal, indicating stable data distribution. However, one feature shows a very high kurtosis value (18.54), suggesting the presence of strong outliers that should be analyzed further.